

Riccardo Talami, Ph.D. M.Sc. B.Arch.

Research Project Manager and Research Scientist - Sustainable Built Environment program

Nanyang Technological University (NTU) Singapore, Energy Research Institute

Website: <https://riccardotalami.github.io/rt/#/>

E-mail: riccardo.talami12@gmail.com

Telephone: +65 8044 6551

Italian citizen and Singapore Permanent Resident (PR)

Research expertise and direction

Riccardo's expertise is positioned at the intersection of *architecture, engineering, and computer science*. Riccardo's research is centred on *data-driven* and *physics-informed building and urban design optimization*.

Riccardo specializes in 1) DIGITAL TRANSFORMATION: developing scalable *data-driven methods, workflows, frameworks, software and tools* that automate performance-based building and urban design processes, and 2) TECHNOLOGICAL TRANSFORMATION: developing *technologies and products* that accelerate the transition to passive and smart built environment.

Riccardo's multi-disciplinary approach integrates *computational and algorithmic thinking, multi-scale physics-informed modelling and mathematical optimization, powered by artificial intelligence* to tackle the challenges of complex built environments at multiple scales: human/individual, room, building, district, and city scale.

His work empowers building and urban stakeholders to shape sustainable, comfortable, smart, efficient, adaptive, and climate-resilient built environments.

Interests

Performance-based Building and Urban Design (*Design + AI + Optimization*)

- Building Performance Modelling and Simulation
- Computational Building and Urban Design
- Building and Urban Design Optimization
- Design Exploration and Knowledge Generation
- Design for Uncertainty and Resilience
- Decision-support during Design Process

Sustainable and Integrated Building and Urban Design (*Design + Technology*)

- Passive Design Strategies and technologies
- Smart Built Environments and sensing technologies
- Urban Microclimate

Education

2019 – 2022

Loughborough, United Kingdom

Doctor of Philosophy (PhD) in Building Science/Building and Architectural Engineering (Building Performance Optimization)

Loughborough University, School of Architecture, Building and Civil Engineering

Thesis title: The sequential design optimization of building performance.

Supervisors: Prof. Jonathan Wright and Dr. Bianca Howard

2014 - 2017

Venice, Italy and Berkeley, California

Master of Science in Architecture and Innovation (Sustainable Design)

University of California Berkely (UC Berkeley) and Università IUAV di Venezia (IUAV), Architecture and Innovation

Thesis title: Recent trends in radiant system technology for heating and cooling

Supervisors: Fred Bauman, PE (UC Berkeley), Prof. Piercarlo Romagnoni (IUAV) and Simone Cappelletti (STEAM Engineering)

2011 - 2014

Venice, Italy

Bachelor of Architecture

Università IUAV di Venezia (IUAV), Architectural Science

Professional experience

- **Research Project Manager and Research Scientist** (2025 - current)

Energy Research Institute (ERI@N)

Nanyang Technological University (Singapore)

Duties: Planning, Leading and managing the research initiatives of the Sustainable Built Environment programme, grants writing and funding acquisition, leading a team of researchers, students and staff supervision, promoting research across government, industry and academia, outreach and generating lead demand.

- **Research Fellow** (2024 - 2025)

Department of Architecture, College of Design and Engineering

National University of Singapore - NUS and Singapore-ETH Centre (Singapore)

Duties: Led and collaborated on research projects related to Performance-based Design exploration and AI, Multi-criteria Decision Making and Results Interpretability, Robust

Building Design Optimization, Urban Building Energy Modelling, Whole life carbon of Urban Stock. Developed and published scientific papers.

- **Research Fellow (2022 - 2024)**

*Department of the Built Environment, College of Design and Engineering
National University of Singapore - NUS (Singapore)*

Duties: Led a team for research projects on Temperature Setpoints and Indoor Environmental Quality, and the development of online tools for practical applications. Collaborated on research projects related to Innovative Building Envelope Systems. Supervised final year students for research thesis and visiting scholars. Developed products and published scientific papers. Managed the procurement for project expenses.

- **Doctoral Researcher (2019 - 2022)**

*School of Architecture, Building and Civil Engineering
Loughborough University (Loughborough, United Kingdom)*

Duties: Researched on Building Performance Optimization. Developed and published scientific papers. Delivered lectures.

- **Co-Instructor and Teaching Assistant (2017 - 2018)**

*ASD - Architecture and Sustainable Design Pillar
Singapore University of Technology and Design - SUTD (Singapore)*

Duties: Developed coursework structure, teaching materials, and assignments. Delivered lectures, conducted weekly reviews and supervised final exams.

- **Research Assistant (2017 - 2018)**

*Design for Climate and Comfort Lab (DCC)
Singapore University of Technology and Design - SUTD (Singapore)*

Duties: Developed and conducted independent research on Radiant Cooling Systems in the Tropics and Human Behaviour in office settings. Collaborated on research projects related to Daylighting in Buildings and Building Performance of Tropical Building Typologies. Developed and published scientific papers.

- **Student Researcher (2016)**

*Center for the Built Environment (CBE)
University of California Berkely - UC Berkeley (United States)*

Duties: Researched on Radiant Cooling Systems, published scientific reports and the development of an online tool.

Teaching Experience

Courses

- BPS5229 Data Science for the Built Environment
Role: **Co-Lecturer** (2025)
National University of Singapore - NUS (Singapore)
- Department of Architecture
Role: **M.Arch. Dissertation Invited Guest Reviewer** (2024)
National University of Singapore - NUS (Singapore)
- Department of Built Environment
Role: **M.Sc. "Passive Systems" course Invited Guest Reviewer** (2024)
National University of Singapore - NUS (Singapore)
- Center for the Energy Resilience and the Built Environment (ERBE)
Role: **Invited Lecturer** (2021)
Loughborough University (United Kingdom)
- Center for the Energy Resilience and the Built Environment (ERBE)
Role: **Invited Lecturer** (2020)
Loughborough University (United Kingdom)
- Department of Architecture and Sustainable Design
Role: **M.Arch. Dissertation Invited Guest Reviewer** (2024)
Singapore University of Technology and Design - SUTD (Singapore)
- 20.112 Sustainable Design Option Studio 2
Role: **Teaching Assistant** (2018)
Singapore University of Technology and Design - SUTD (Singapore)
- 20.112 Sustainable Design Option Studio 3
Role: **Teaching Assistant** (2018)
Singapore University of Technology and Design - SUTD (Singapore)
- 20.111 Sustainable Design Option Studio 1
Role: **Teaching Assistant** (2018)
Singapore University of Technology and Design - SUTD (Singapore)
- 20.223 History, Theory and Culture 3: Contemporary Architecture Between Technology, Science and Culture
Role: **Co-Lecturer** (2017)
Singapore University of Technology and Design - SUTD (Singapore)

Supervision and mentoring

- Tran Le Vu Research Fellow (2026 - current)
- Nima Bonyadi PhD candidate (2024 - current)
- Wanyu Pei PhD candidate (2024 -2025)
- Pradeep Alva PhD candidate (2024 - 2025)
- Pegah Eshraghi PhD candidate (2024 - 2025)
- Xudong Jia Master Student (2023 - 2024)

- Ilyas Dawoodjee Research Engineer (2023 - 2024)
- Xinhao Hu Visiting PhD Candidate (2023 - 2024)
- Glenda Cheng Undergraduate Dissertation (2023 - 2024)
- Sabrina Tay Undergraduate Dissertation (2023)
- Shirlynn Koh Undergraduate Dissertation (2023)
- Thomas Firsich Visiting Master Student (2023)

Guest Lectures

- **ASEAN-China Symposium on Built Environment Renewal and Sustainable Energy Utilization**
Title: Cooling Singapore through Passive Design (2025)
Tianjin University, Guizhou University, Guiyang University (China)
- **Center for the Energy Resilience and the Built Environment (ERBE)**
Title: Performance-based building design in visual coding environments (2021)
Loughborough University (United Kingdom)
- **Center for the Energy Resilience and the Built Environment (ERBE)**
Title: The role of Parametric Design, Building Performance Simulation and Optimization in Performance-based building design (2020)
Loughborough University (United Kingdom)

Research grants

External Research Grants

\$2.7 M in pending grant proposals (400k awarding in process)

- **(awarding process) Nemetschek foundation** proposal on AI-enhanced climate-responsive microclimate real-time insights on urban heat island (Co-PI, \$400k, 48 months).
- **(pending) Cities of Tomorrow 2025** proposal on natural ventilation for climate-adaptive design and resilience (PI, \$495.3k, 18 months).
- **(pending) GBIC 2025** proposal on natural ventilation in hospital wards (PI, \$742.3k, 24 months)
- **(pending) YIRG 2025** proposal on cooling panels design (PI, \$325k, 24 months).
- **(pending) GBIC 2025** proposal on thermoelectric cooling (Co-PI, \$382k, 24 months)
- **(pending) NTUitive 2026** proposal on climate-adaptive cooling panel design (PI, \$250k, 18 months)
- **(pending) Temasek Foundation 2026** proposal on climate-adaptive cooling system panel (PI, \$550k, 18 months)

Internal Research Grants

\$ 540 k in grants managed

- Energy retrofit and digitalization of EPGC building to achieve GreenMark Platinum SLE (\$500k – \$125k used so far for ACMV retrofit, 2025 - current).

- Active phase change material cooling system demonstration space (\$40k, 2025 - current).

Grant management and coordination

\$ 454 k in grants managed

- AI and IoT Enhanced Sustainable Smart Cooling Platform for Optimal Comfort and Energy Efficiency (JTC DECAL, \$454k, 2025 - current).

Academic Service

Reviewing

- **Reviewer for Journal of Asian Architecture and Building Engineering, Taylor and Francis** (Jan 2026 – Current).
- **Reviewer for Frontiers in Artificial Intelligence, Frontiers** (November 2025 – Current).
- **Reviewer for Energy and Buildings journal, Elsevier** (August 2025 – Current).
- **Reviewer for 2025 Sustainable Asia Conference** (August 2025 – Current)
- **Reviewer for CISBAT 2025 International Conference** (February 2025 – Current)
- **Reviewer for Journal of Architectural Engineering** (2024 - Current)
- **Reviewer for Building Simulation 2023: 18th Conference of IBPSA** (August 2022 - September 2023)
4-6 September 2023, Shanghai (China) - Hybrid
- **Reviewer for Building Simulation 2021: 17th Conference of IBPSA** (August 2020 - September 2021)
1-3 September 2021, Bruges (Belgium)
- **Reviewer for Building Simulation and Optimization 2020** (September 2019 - September 2020)
21-22 September 2020, Loughborough (UK)
- **Reviewer for Building Simulation 2019: 16th Conference of IBPSA** (August 2018 - September 2019)
2-4 September 2019, Rome (Italy)

Associations and Memberships

- **Committee Member - Electronic Communications - International Building Performance Simulation Association (IBPSA-Singapore)** (2024 - Current)
- **Professional Committee Member - International Building Performance Simulation Association (IBPSA-Singapore)** (2024 - Current)
- **Student Member - International Building Performance Simulation Association (IBPSA-United Kingdom)** (2019 - 2022)

Conference organisation

- **Scientific Committee – 6th Asian Conference of the International Building Performance Simulation Association (Asim) (IBPSA) (2026)**

Presentations

- **Cooling Singapore through Passive Design (2025)**
ASEAN-China Symposium on Built Environment Renewal and Sustainable Energy Utilization (China)
- **Hierarchical decomposition approaches for building design (2025)**
BUDS Research Lab at the National University of Singapore (Singapore)
- **Performance-based building and urban design (2024)**
BUDS Research Lab at the National University of Singapore (Singapore)
- **Examining Different Placement Strategies for Indoor Environmental Quality Sensors in Office Environments (2024)**
Building Robotics Lab at the National University of Singapore (Singapore)
- **Energy savings from dynamic temperature setpoints and setbacks (2023)**
Funding agency at the National University of Singapore (Singapore)
- **Multi-criteria robustness assessment of a sequential whole-building design optimization (2021)**
IBPSA Conference in Bruges (Belgium)
- **A comparison between sequential and simultaneous whole-building design optimization for building performance (2020)**
IBPSA - UK Conference in Loughborough (UK)
- **The sequential design optimization of building performance (2020)**
Funding agency at Loughborough University (UK)
- **Sensitivity of design parameters on energy, system and comfort performances for radiant cooled office buildings in the tropics. (2019)**
IBPSA Conference in Rome (Italy)
- **The sequential design optimization of building performance (2019)**
Doctoral Seminar at Loughborough University (UK)

Events and Training

Conferences

- **International Conference on Resilient Systems**
28-30 August 2024, Singapore
- **Building Simulation 2023: 18th Conference of IBPSA**
4-6 September 2023, Shanghai (China)
- **Building Simulation 2021: 17th Conference of IBPSA**
1-3 September 2021, Bruges (Belgium)
- **Building Simulation and Optimization (BSO) 2020**

21-22 September 2020, Loughborough (UK)

- **Building Simulation 2019: 16th Conference of IBPSA**
2-4 September 2019, Rome (Italy)

Symposiums

- **ASEAN-China Symposium on Built Environment Renewal and Sustainable Energy Utilization**
26-28 July 2025, Tianjin University, Guizhou University, Guiyang University (China)

Workshops

- **Grasshopper Level 2 course from McNeel (2021)**
Online
- **Grasshopper Level 1 course from McNeel (2021)**
Online
- **Teaching - advanced module 2 (2021)**
Loughborough University (UK)
- **Teaching - basic module 1 (2020)**
Loughborough University (UK)
- **Certified Peer Reviewer Course from Elsevier (2020)**
Online
- **Exploring Structures and Features of Research Articles (2019)**
Loughborough University (UK)
- **Exploring Research Article Abstracts (2019)**
Loughborough University (UK)
- **Exploring Research Article Introduction (2019)**
Loughborough University (UK)
- **Exploring Research Article Methods Sections (2019)**
Loughborough University (UK)
- **Exploring Research Article Discussion Sections (2019)**
Loughborough University (UK)
- **Exploring Research Article Methods Sections (2019)**
Loughborough University (UK)
- **Exploring Research Article Results Sections (2019)**
Loughborough University (UK)
- **Finding resources for your literature review and beyond (2019)**
Loughborough University (UK)

Exhibitions

- **Asia Clean Energy Summit 2025 exhibition (2025)**
Marina Bay Sands Convention Centre (Singapore)
- **Doctoral College Summer Showcase (2020)**
Loughborough University (UK)

- **Research conference 2019 – Visibility and Impact (2019)**
Loughborough University (UK)
- **Doctoral College Summer Showcase (2019)**
Loughborough University (UK)

Projects

Active (11)

2025 - Current

Singapore

AI and IoT Enhanced Sustainable Smart Cooling Platform for Optimal Comfort and Energy Efficiency

Energy Research Institute, Nanyang Technological University (NTU)

The research enhances GAIATech's HyperCool™ and iVAV™ platform, an integrated ACMV optimization solution, with additional integration and control capabilities.

Role: Project Coordinator

2025 - Current

Singapore

Enhancing the cooling potential of green walls

Energy Research Institute, Nanyang Technological University (NTU)

The research investigates the enhancement of cooling potential and thermal comfort of conventional green walls when paired with phase change materials and hydrogels.

Role: Project Leader

2025 - Current

Singapore

Viability of free-running buildings in the tropics

Energy Research Institute, Nanyang Technological University (NTU)

The research explores the viability and performance of free-running buildings in the tropics and proposes a framework and tool to maximize available technologies to achieve super-low energy pathways.

Role: Project Leader

2025 - Current

Singapore

Sustainable active cooling via phase change materials

Energy Research Institute, Nanyang Technological University (NTU)

The research develops and evaluates the cooling potential of walls and columns activated via phase change material integration for indoor thermal comfort.

Role: Project Coordinator

2025 - Current

Singapore

Synthetic data generation for Urban Building Energy Modelling

Energy Research Institute, Nanyang Technological University (NTU)

The research develops a fast and accurate framework to generate synthetic data based on building codes and certifications for modelling and simulation of urban districts.

Role: Project Leader

2025 - Current

Singapore

Retrofit of EPGC building for Platinum Super Low Energy Green Mark

Energy Research Institute, Nanyang Technological University (NTU)

The research transforms EPGC building in Jurong Island into a smart and energy-efficient facility for demonstration projects, achieving a Platinum Super Low Energy (SLE) Green Mark certification.

Role: Project Coordinator

2024 - Current

Singapore

AI-driven frameworks for Computational Building Performance Design

Singapore - ETH Centre, National University of Singapore (NUS)

The research develops computational frameworks for multi-criteria decision making and quick design explorations in building performance design.

Role: Project Leader

In Collaboration with: *Dr. Alberto Costa, Senior Researcher at Singapore-ETH Centre, Prof. Rudi Stouffs, Assoc. Professor at NUS, Prof. Clayton Miller, Assoc. Professor at NUS.*

2024 - Current

Singapore

Problem decomposition approaches in Building Design

Concordia University (Canada), National University of Singapore (NUS), Singapore-ETH Centre

The research investigates the application of decomposition approaches in building design to reduce problem complexity.

Role: Collaborator

In Collaboration with: *Nima Bonyadi, PhD Research Assistant at Concordia University.*

2024 - Current

Singapore

The role of Archetypes in Urban Building Energy Modelling

Singapore - ETH Centre, National University of Singapore (NUS)

The research improves accuracy and detail of conventional archetypes for urban performance simulation studies.

Role: Project Leader

In Collaboration with: *Wanyu Pei, PhD student at NUS and Researcher at Singapore-ETH Centre.*

2024 - Current

Singapore

The impact of thermal inertia in mitigating energy disruptions

Singapore - ETH Centre, National University of Singapore (NUS)

The research investigates the potential of building thermal mass to mitigate urban energy disruptions.

Role: Collaborator

In Collaboration with: *Andrea Bartolini, Postdoctoral Researcher at Singapore-ETH Centre.*

2024 - Current

Singapore

Whole life carbon assessment of urban building stock

Singapore - ETH Centre, National University of Singapore (NUS)

The research proposes a framework for conducting city-scale WLCA, integrating both embodied and operational carbon assessments to evaluate the circularity potential of urban building stock.

Role: Collaborator

In Collaboration with: *Pradeep Alva, PhD student at NUS and Researcher at Singapore-ETH Centre.*

Completed (10)

2024 - 2025

Singapore

AI-driven frameworks for urban environments

Singapore - ETH Centre, National University of Singapore (NUS)

The research enhances Urban Building Energy Modelling and develops frameworks for urban natural and built environments based on their performance.

Role: Collaborator

In Collaboration with: *Pegah Eshraghi, PhD student at Shahid Beheshti University.*

2023 - 2024

Singapore

Exploring novel cooling technologies for building façades

National University of Singapore (NUS)

The research provides a framework for the synergistic integration of vertical greenery systems (VGS), heat-reflective paints, and façade shading systems.

Role: Collaborator

In Collaboration with: *Iqbal Shah, PhD student at NUS.*

2023 - 2024

Singapore

Human-centric Indoor Environmental Quality (IEQ) sensing

National University of Singapore (NUS)

The research identifies IEQ sensor placements that most accurately reflect the environmental conditions experienced by occupants in office settings.

Role: Project Leader

2023 - 2024

Singapore

Optimizing HVAC temperature setpoints for energy efficiency

National University of Singapore (NUS)

The research evaluates the energy-saving potential of dynamically adjusting room temperature setpoints and setbacks of HVAC systems based on varying outdoor weather and indoor occupancy conditions.

Role: Project Leader

2019 - 2022

Loughborough, United Kingdom

The sequential design optimization of building performance

Loughborough University

The research proposes and evaluates effectiveness, reliability, and efficiency of a sequential approach for multi-objective design optimization of building geometry, fabric, HVAC system and controls.

Role: Project Leader

In Collaboration with: *Prof. Jonathan Wright, Loughborough University, and Dr. Bianca Howard, Columbia University.*

2017 - 2018

Singapore

Comparing laboratory and field studies of occupant lighting experience

Singapore University of Technology and Design (SUTD)

The research investigates if laboratory conclusions related to visual discomfort and lighting quality are applicable to real building conditions by comparing 40 laboratory post-occupancy evaluation responses to 40 field responses in actual office workspaces in Singapore.

Role: Collaborator

In Collaboration with: *Dr. J. Alstan Jakubiec, Assist. Professor at Toronto University.*

2017 - 2018

Singapore

Subjective and measured evidence for residential lighting metrics in the tropics

Singapore University of Technology and Design (SUTD)

The research presents a comprehensive study methodology to craft statistically valid subjective models based on predictive lighting simulation data, comparing measured and simulated lighting levels in 17 residential housing units in Singapore against the subjective opinions of 35 participants who reside in the units.

Role: Collaborator

In Collaboration with: *Dr. J. Alstan Jakubiec, Assist. Professor at Toronto University.*

2017 - 2018

Singapore

Radiant Cooling Systems in the Tropics

Singapore University of Technology and Design (SUTD)

The research supports early-design decisions when integrating radiant cooling systems in architectural design by identifying key design parameters influencing energy consumption, system operation, and thermal comfort in the tropics.

Role: Project Leader

In Collaboration with: *Dr. J. Alstan Jakubiec, Assist. Professor at Toronto University.*

2017 - 2018

Singapore

Sustainable Futures: Cooling

Singapore University of Technology and Design (SUTD)

The project investigates human behaviors through long-term observational field studies in commercial and residential spaces, analyzing air-conditioning, window shading, and lighting use. The findings inform predictive models and influence building design decisions.

Role: Collaborator

In Collaboration with: *Dr. J. Alstan Jakubiec, Assist. Professor at Toronto University.*

2016

California, United States

Optimizing Radiant Systems for Energy Efficiency and Comfort

University of California, Berkeley (UC Berkeley) - Center for the Built Environment

This project develops a database of over 400 commercial buildings globally, displayed on an online interactive map.

Role: Collaborator

In Collaboration with: Fred Bauman, PE

Core Skills

Transferrable Skills

- Excellent journal, conference paper, and technical report writing skills
- Ability to work in a team and independently
- Excellent presentation and communication skills (verbal and written)
- Ability to work in a multidisciplinary and multicultural environment
- Ability to work on several projects simultaneously, with clear deadlines and under pressure

Software and Technical

- Energy Analysis: Energy Plus (Conventional, Design Builder and Open Studio Interfaces), ArchSim and Honeybee plug-ins for Grasshopper
- Environmental and Micro-climate Analysis: ENVI-met

- Energy Certification: Master Clima 11300
- Solar and Environmental Analysis: Ecotect, Climate Consultant
- Daylight Analysis: Radiance (DIVA and Ladybug interfaces)
- Parametric Modeling: Rhinoceros/Grasshopper
- Statistics and Data Science: R
- Programming: Python, Java
- Environmental Sensors: HOBO products, DustTrak™ DRX Aerosol Monitor 8534, XL2 Audio and Acoustic Analyzer, VelociCalc Multi-Function Ventilation Meter 9565, RAE Systems ppbRAE 3000+ Portable Handheld VOC Monitor, Testo Luxmeter
- Microcontrollers: Raspberry Pi, Arduino
- Architecture/Engineering Drafting and Modeling: AutoCAD, ArchiCAD, Revit, SketchUp, Rhinoceros
- Graphic and Editing: Photoshop, Illustrator, InDesign
- General: Microsoft Office package

Certifications/Courses

- Certified Peer Reviewer Course from Elsevier, January 8-9, 2020
- Grasshopper Level 1 course from McNeel, 12 hours, January 25-29, 2021
- Grasshopper Level 2 course from McNeel, 18 hours, January 25-29, 2021

Languages

- English: Full Professional Proficiency
- Italian: Native Proficiency
- French: Elementary Proficiency
- Spanish: Elementary Proficiency

Awards

- **UK Engineering and Physical Sciences Research Council Scholarship**, Engineering and Physical Sciences Research Council (2019 - 2022)
- **Winner of Best Poster Award** for “Subjective and Measured Evidence for Residential Lighting Metrics in the Tropics” (in collaboration with Jakubiec, J. Alstan; Srisamranrungruang, Thanyalak; Kong, Zhe; Quek, Geraldine), 16th International IBPSA Conference (2019).

Research outputs

- Journal papers: 9, Conference papers: 5, Reports: 1, Thesis: 2.
- Upcoming publications: Undergoing peer-review: 4, In process/Writing-up: 5
- Products: 2
- Software: 2

Publications

Publications are available at: <https://scholar.google.com/citations?user=EmFyzowAAAAJ&hl=en>.

Peer-reviewed Journals

- Eshraghi, P., Dehnavi, A. N., Mirdamadi, M., **Talami, R.**, and Zomorodian Z. S. 2025. An AI-driven framework for rapid and localized optimizations of urban open spaces. *Smart and Sustainable Built Environment*, 1-27.
- Bonyadi, N., **Talami, R.**, and Lee. B. 2025. Breaking-down Building Design Problems with Decomposition Approaches: A review. *Energy and Built Environment*, In press.
- Eshraghi, P., **Talami, R.**, Dehnavi, A. N., Mirdamadi, M. and Zomorodian Z. S. 2025. Adopting Explainable-AI to investigate the impact of urban morphology design on energy and environmental performance in dry-arid climates. *Advances in Building Energy Research*, 19, 497-531.
- **Talami, R.**, Hu, X., Dawoodjee, I. and Ghahramani, A. 2025. Examining Different Placement Strategies for Indoor Environmental Quality Sensors in Office Environments. *Science and Technology for the Built Environment*, 31, 786-805
- **Talami, R.**, Wright, J., and Howard, B. 2025. Evaluating the effectiveness, reliability and efficiency of a multi-objective sequential optimization approach for building performance design. *Energy and Buildings*, 329, 115242.
- Shah, I., Su, X., **Talami, R.** and Ghahramani, A. 2024. Enhancing Building Envelopes: Parametric Analysis of Shading Systems for Opaque Facades and Their Comparison with Cool Paints. *Energy and Built Environment*, 6(5), 904-919.
- **Talami, R.**, Dawoodjee, I. and Ghahramani, A., 2024. Demystifying energy savings from dynamic temperature setpoints under weather and occupancy variability. *Energy and Built Environment*, 5(6), 878-888
- **Talami, R.**, Dawoodjee, I. and Ghahramani, A., 2023. Quantifying Energy Savings from Optimal Selection of HVAC Temperature Setpoints and Setbacks across Diverse Occupancy Rates and Patterns. *Buildings*, 13(12), 2998
- **Talami, R.** and Jakubiec, J.A., 2020. Early-design sensitivity of radiant cooled office buildings in the tropics for building performance. *Energy and Buildings*, 223, 110177

Peer-reviewed Conferences

- Alva, P., **Talami, R.**, Pei, W., Miller, C. and Stouffs, R., 2025. A framework for building material recycling potential: Whole-life carbon assessment for residential buildings at city-scale. *Journal of Physics: Conference Series*, 3140, 132008.
- **Talami, R.**, Wright, J. and Howard, B., 2021. Multi-criteria robustness assessment of a sequential whole-building design optimization. In *Proceedings of the 17th IBPSA International conference, 1-3 September 2021, Bruges (Belgium)*.
- **Talami, R.**, Wright, J. and Howard, B., 2020. A comparison between sequential and simultaneous whole-building design optimization for building performance. In *Proceedings of the Building Simulation and Optimization Conference, Loughborough, UK, September 21-22*.

- **Talami, R.** and Alstan Jakubiec, J., 2019, September. Sensitivity of design parameters on energy, system and comfort performances for radiant cooled office buildings in the tropics. In *Proceedings of the 16th IBPSA International conference, 2-4 September 2019, Rome (Italy)*.
- Jakubiec, J.A., Srisamranrungruang, T., Kong, Z., Quek, G. and **Talami, R.**, 2019. Subjective and measured evidence for residential lighting metrics in the tropics. *Proceedings of the 16th IBPSA International conference, 2-4 September 2019, Rome (Italy)*.

Thesis

- **Talami, R.**, 2022. The sequential design optimization of building performance. (Doctoral dissertation, Loughborough University).
- **Talami, R.**, 2017. Recent trends in hydronic radiant system technology for heating and cooling. (Master dissertation, University of California Berkeley/IUAV University of Venice).

Reports

- **Talami, R.**, Karmann, C., Bauman, F., Schiavon, S. and Raftery, P., 2017. Recent trends in radiant system technology in North America (UC Berkeley online repository).

Upcoming (undergoing peer-review or in-process)

- Pei, W., **Talami, R.**, Miller, C., Stouffs, R. 2026. Dynamic Micro-Archetype Modelling: Towards Parametric and Simulation-Driven Archetype Development for Urban Energy Stock Estimation.
- **Talami, R.**, Costa, A., Miller, C., Stouffs, R. 2026. A Framework for Fast, Interpretable and Uncertainty-aware Decision-Support for Building Performance Design.
- Alva, P., **Talami, R.**, Pei, W., Sibenik, G., Mosteiro-Romero, M., Miller, C., Stouffs, R. 2026. Potential recyclable materials in buildings: A framework for greenhouse gas emissions assessment of residential buildings in Singapore
- Costa, A., and **Talami, R.** 2026. INFERNO: A Framework for Knowledge Generation with Expensive Sample Evaluations.
- **Talami, R.**, Alva, P., Pei, W., Miller, C., Stouffs, R. 2026. Inference-based framework for synthetic data generation and enrichment of urban building stock.
- **Talami, R.** and Zulkefli, F. 2026. A Techno-economic Framework to aid Decision-making towards Passive and Free-Running Buildings.
- **Talami, R.**, Hertel, V., Zulkefli, F. 2026. From air-cooled to naturally ventilated built environment: a perspective.
- **Talami, R.** and Zulkefli, F. 2026. Enhancing the cooling and outdoor thermal comfort potential of green walls via phase change materials and hydrogels.
- **Talami, R.** and Alstan Jakubiec, J., 2026. Predicting human behaviors through air-conditioning, window shading, and lighting use.

Products

Indoor Environmental Quality sensor package v.2.0 (2023 - 2024)

Designed and assembled 40 Arduino-based sensor packages that monitor 9 Indoor Environmental Quality metrics continuously: Carbon dioxide, Particulate matter, Total volatile organic compound, Illuminance, Temperature, Humidity, and Sound continuously. Prototyped and 3D printed 40 sensor enclosures with reduced dimensions. Introduced fast-charging battery and Wi-Fi connection.

Indoor Environmental Quality sensor package v.1.0 (2023 - 2024)

Designed and assembled 20 Raspberry-pi-based sensor packages that monitor 9 Indoor Environmental Quality metrics continuously: Carbon dioxide, Particulate matter, Total volatile organic compound, Illuminance, Temperature, Humidity, and Sound. Prototyped and 3D printed 20 boltless sensor enclosures for easy assembly and maintenance.

Tools

Optimal Temperature Setpoint Tool (2023 - 2024)

The Optimal Temperature Setpoint Tool allows users to identify the optimal temperature setpoint based on occupancy rates, patterns, and outdoor air temperature values. [\[link\]](#)

CBE Radiant Systems Map (2016)

The CBE Radiant Systems Map displays a database of over 400 commercial buildings using radiant cooling and heating in the world as online interactive map. The tool has obtained 23,600 views so far. [\[link\]](#)